

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
import numpy as np

from tensorflow.keras.layers import Input, Dense

from tensorflow.keras.models import Model

from tensorflow.keras.datasets import mnist

import matplotlib.pyplot as plt


# Load the MNIST dataset

(x_train, _), (x_test, _) = mnist.load_data()

x_train = x_train.astype('float32') / 255.

x_test = x_test.astype('float32') / 255.


# Flatten images for the fully connected layer (28x28 images -> 784-dimensional vectors)

x_train = x_train.reshape((len(x_train), np.prod(x_train.shape[1:])))

x_test = x_test.reshape((len(x_test), np.prod(x_test.shape[1:])))


# Set the dimensions of encoding

encoding_dim = 32 # This is the size of the encoded representations


# Input placeholder

input_img = Input(shape=(784,))


# Encoding layer

encoded = Dense(encoding_dim, activation='relu')(input_img)


# Decoding layer

decoded = Dense(784, activation='sigmoid')(encoded)
```

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```
# Autoencoder model

autoencoder = Model(input_img, decoded)


# Encoder model

encoder = Model(input_img, encoded)


# Decoder model

encoded_input = Input(shape=(encoding_dim,))

decoder_layer = autoencoder.layers[-1]

decoder = Model(encoded_input, decoder_layer(encoded_input))


# Compile the autoencoder

autoencoder.compile(optimizer='adam', loss='binary_crossentropy')


# Train the autoencoder

autoencoder.fit(x_train, x_train,

                epochs=50,

                batch_size=256,

                shuffle=True,

                validation_data=(x_test, x_test))


# Encode and decode some test images

encoded_imgs = encoder.predict(x_test)

decoded_imgs = decoder.predict(encoded_imgs)
```

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```
# Display original and reconstructed images

n = 10 # Number of images to display

plt.figure(figsize=(20, 4))

for i in range(n):

    # Original

    ax = plt.subplot(2, n, i + 1)

    plt.imshow(x_test[i].reshape(28, 28), cmap='gray')

    plt.axis('off')

    # Reconstructed

    ax = plt.subplot(2, n, i + 1 + n)

    plt.imshow(decoded_imgs[i].reshape(28, 28), cmap='gray')

    plt.axis('off')
```

OUTPUT

```
runfile('D:/academic year 2024-2025/AUTOENCODER.py', wdir='D:/academic year 2024-2025')
```

Epoch 1/50

235/235 ————— **3s 7ms/step - loss: 0.3797 - val_loss: 0.1879**

Epoch 2/50

235/235 ————— **1s 6ms/step - loss: 0.1778 - val_loss: 0.1523**

Epoch 3/50

235/235 ————— **2s 7ms/step - loss: 0.1486 - val_loss: 0.1341**

Epoch 4/50

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235/235 ————— 2s 7ms/step - loss: 0.1322 - val_loss:
0.1218

Epoch 5/50

235/235 ————— 2s 7ms/step - loss: 0.1205 - val_loss:
0.1130

Epoch 6/50

235/235 ————— 2s 7ms/step - loss: 0.1127 - val_loss:
0.1072

Epoch 7/50

235/235 ————— 2s 6ms/step - loss: 0.1072 - val_loss:
0.1027

Epoch 8/50

235/235 ————— 2s 7ms/step - loss: 0.1030 - val_loss:
0.0991

Epoch 9/50

235/235 ————— 2s 7ms/step - loss: 0.0998 - val_loss:
0.0968

Epoch 10/50

235/235 ————— 2s 7ms/step - loss: 0.0977 - val_loss:
0.0953

Epoch 11/50

235/235 ————— 2s 8ms/step - loss: 0.0964 - val_loss:
0.0943

Epoch 12/50

235/235 ————— 2s 8ms/step - loss: 0.0956 - val_loss:
0.0937

Epoch 13/50

235/235 ————— 2s 7ms/step - loss: 0.0948 - val_loss:
0.0933

Epoch 14/50

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235/235 ————— 2s 7ms/step - loss: 0.0946 - val_loss: 0.0931

Epoch 15/50

235/235 ————— 2s 7ms/step - loss: 0.0943 - val_loss: 0.0928

Epoch 16/50

235/235 ————— 2s 7ms/step - loss: 0.0940 - val_loss: 0.0927

Epoch 17/50

235/235 ————— 2s 7ms/step - loss: 0.0940 - val_loss: 0.0925

Epoch 18/50

235/235 ————— 1s 6ms/step - loss: 0.0938 - val_loss: 0.0924

Epoch 19/50

235/235 ————— 1s 6ms/step - loss: 0.0937 - val_loss: 0.0923

Epoch 20/50

235/235 ————— 1s 6ms/step - loss: 0.0935 - val_loss: 0.0922

Epoch 21/50

235/235 ————— 1s 6ms/step - loss: 0.0934 - val_loss: 0.0921

Epoch 22/50

235/235 ————— 1s 6ms/step - loss: 0.0937 - val_loss: 0.0922

Epoch 23/50

235/235 ————— 1s 5ms/step - loss: 0.0935 - val_loss: 0.0921

Epoch 24/50

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235/235 ————— 1s 6ms/step - loss: 0.0934 - val_loss:
0.0920

Epoch 25/50

235/235 ————— 1s 6ms/step - loss: 0.0932 - val_loss:
0.0919

Epoch 26/50

235/235 ————— 1s 6ms/step - loss: 0.0932 - val_loss:
0.0919

Epoch 27/50

235/235 ————— 1s 6ms/step - loss: 0.0933 - val_loss:
0.0918

Epoch 28/50

235/235 ————— 1s 6ms/step - loss: 0.0930 - val_loss:
0.0919

Epoch 29/50

235/235 ————— 1s 6ms/step - loss: 0.0931 - val_loss:
0.0918

Epoch 30/50

235/235 ————— 1s 6ms/step - loss: 0.0932 - val_loss:
0.0918

Epoch 31/50

235/235 ————— 1s 6ms/step - loss: 0.0930 - val_loss:
0.0918

Epoch 32/50

235/235 ————— 1s 6ms/step - loss: 0.0929 - val_loss:
0.0918

Epoch 33/50

235/235 ————— 1s 6ms/step - loss: 0.0929 - val_loss:
0.0918

Epoch 34/50

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235/235 ————— 1s 6ms/step - loss: 0.0929 - val_loss:
0.0918

Epoch 35/50

235/235 ————— 1s 6ms/step - loss: 0.0931 - val_loss:
0.0917

Epoch 36/50

235/235 ————— 1s 6ms/step - loss: 0.0928 - val_loss:
0.0916

Epoch 37/50

235/235 ————— 1s 6ms/step - loss: 0.0928 - val_loss:
0.0917

Epoch 38/50

235/235 ————— 1s 6ms/step - loss: 0.0927 - val_loss:
0.0917

Epoch 39/50

235/235 ————— 1s 6ms/step - loss: 0.0928 - val_loss:
0.0917

Epoch 40/50

235/235 ————— 1s 6ms/step - loss: 0.0926 - val_loss:
0.0919

Epoch 41/50

235/235 ————— 1s 6ms/step - loss: 0.0927 - val_loss:
0.0916

Epoch 42/50

235/235 ————— 1s 6ms/step - loss: 0.0928 - val_loss:
0.0916

Epoch 43/50

235/235 ————— 1s 6ms/step - loss: 0.0927 - val_loss:
0.0915

Epoch 44/50

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235/235 ————— 1s 6ms/step - loss: 0.0927 - val_loss: 0.0916

Epoch 45/50

235/235 ————— 1s 6ms/step - loss: 0.0927 - val_loss: 0.0916

Epoch 46/50

235/235 ————— 1s 6ms/step - loss: 0.0930 - val_loss: 0.0915

Epoch 47/50

235/235 ————— 2s 6ms/step - loss: 0.0927 - val_loss: 0.0916

Epoch 48/50

235/235 ————— 1s 6ms/step - loss: 0.0929 - val_loss: 0.0916

Epoch 49/50

235/235 ————— 1s 6ms/step - loss: 0.0926 - val_loss: 0.0916

Epoch 50/50

235/235 ————— 1s 6ms/step - loss: 0.0928 - val_loss: 0.0916

313/313 ————— 1s 1ms/step

313/313 ————— 1s 2ms/step

